THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 6.0 V	4 x 1.5.
2	**(a)** using damaged wires	Damaged wires are dangerous.
3	**(d)** becomes brighter	More voltage.
4	**(a)** 0 V	They cancel.
5	**(d)** are insulators	Block current.
6	**(c)** a cell, bulb and wires	Must have a source and indicator.
7	**(a)** the filament is broken or the circuit is open	Gap in path.
8	**(a)** each bulb terminal connected to a different cell terminal	Complete path.
9	**(b)** glow	Graphite conducts.
10	**(a)** the switch is closed	Closed circuit.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Coin is a conductor, completing the circuit (1); Plastic is an insulator (1).	1+1
12	Voltages cancel (1); Bulb does not glow (1).	1+1
13	Switch off the main supply (1); Use the same rating (1).	1+1
14	Long wire fits in a small space (1); More heat in a small region (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total voltage 6 V (1); Matches bulb rating (1); Bulb glows normally (1).	1+1+1
16	Senses large current (1); Switch trips (1); Can be reset (1).	1+1+1
17	Switch off (1); Hold the plug, not the wire (1); Dry hands (1).	1+1+1
	OR: Cell, bulb, wires and two free ends (1); Touch objects across ends (1); Conductors make the bulb glow (1).	1+1+1
18	Geysers need more current (1); Thick wires carry it safely (1); Thin wires would overheat (1).	1+1+1
	OR: Soft and marks paper (1); Conducts electricity (1); A non-metal conductor (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cell, bulb, wires, free ends (1); Place samples: iron nail, coin, eraser, plastic comb, graphite (1); Bulb glows for conductors (1); Table of results (1); Conclude metals and graphite conduct (1).	1+1+1+1+1
	OR: Switch off main supply (1); Use insulated tools (1); Wear rubber gloves (1); Check with a tester (1); Stand on dry insulated mat (1).	1+1+1+1+1
20	Connect nichrome wire to cells (1); Pass current (1); Wire gets hot (1); Increase cells: hotter (1); Warn about burns (1).	1+1+1+1+1
	OR: The two wires of the circuit touch (1); Very large current flows (1); Wires get very hot (1); Sparks and fire (1); Fuse or MCB cuts off (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
- Writing that the fuse is connected in parallel with the appliance; it is connected in **series**.
- Mixing up the heating effect (heat from current) with the cause of the glow of a bulb; the filament glows because it becomes very hot.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** more cells are added in series	More voltage.
2	**(d)** light and a good conductor	Low cost.
3	**(c)** copper and aluminium	Good conductors.
4	**(b)** the other goes off too	Break in the single path.
5	**(d)** a line joining two points	Closed path.
6	**(a)** copper wire	Conductor.
7	**(b)** laptop	Heat is waste.
8	**(a)** a lower	So it melts first.
9	**(a)** salt solution	Contains ions.
10	**(c)** waste much energy as heat	Replaced by LEDs.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	LED emits light directly without heating a filament (1); It wastes less energy (1).	1+1
12	No complete path through them to the ground (1); They touch only one wire (1).	1+1
13	A thin wire carries current less easily (1); More heat is produced in a thin wire (1).	1+1
14	Saves energy (1); Reduces hazards (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cells are opposed (1); Voltage cancels (1); No current in the bulb (1).	1+1+1
16	Add cells in series (1); Increase voltage (1); More current (1).	1+1+1
17	Cell and connections (1); Filament (1); Switch (1).	1+1+1
	OR: Overloading draws large current (1); Fuse wire melts (1); Circuit breaks (1).	1+1+1
18	Short circuit risk (1); Shock risk (1); Fire risk (1).	1+1+1
	OR: Fuse is protective (1); Switch is for control (1); Fuse melts (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fuse melts on overload (1); MCB trips (1); Wrong fuse does not melt (1); Wires overheat (1); Fire risk (1).	1+1+1+1+1
	OR: Length (1); Thickness (1); Material (1); Amount of current (1); Time (1).	1+1+1+1+1
20	Cells, bulb, switch, reflector (1); Closed switch completes circuit (1); Dead cells (1); Broken filament (1); Loose contacts (1).	1+1+1+1+1
	OR: Main switch and fuse or MCB at the entry (1); Meter records the energy used (1); Wires carry electricity to sockets (1); Plug connects the appliance to the socket (1); Switch controls the appliance (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
- Writing that the fuse is connected in parallel with the appliance; it is connected in **series**.
- Mixing up the heating effect (heat from current) with the cause of the glow of a bulb; the filament glows because it becomes very hot.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** fit a long wire in a small space	Compact heating.
2	**(c)** distilled water	Pure water is a poor conductor.
3	**(b)** proper insulation of wires	Insulation.
4	**(b)** series	One path.
5	**(a)** immersion heater	Heater.
6	**(a)** switch off the supply and remove the plug	Disconnect from supply.
7	**(d)** burning smell	Warning.
8	**(a)** it will not break the circuit on overload	No protection.
9	**(a)** the current heats it until it glows	Electrical energy converts to heat and light.
10	**(b)** the whole current passes through it	Safety.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Wet manja can conduct (1); Can cause a shock (1).	1+1
12	Circuit breaks (1); No current flows (1).	1+1
13	Has a low melting point (1); Melts to protect (1).	1+1
14	Alloy gets hot without melting (1); Alloy does not rust easily (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Metals conduct (1); Plastic insulates (1); Both together are safe (1).	1+1+1
16	Low power use (1); Long life (1); Little heat (1).	1+1+1
17	Water conducts (1); Short circuits (1); Shock (1).	1+1+1
	OR: Current heats wires (1); Energy wasted as heat (1); Cooling fans are used (1).	1+1+1
18	Opens or closes the path (1); Controls current (1); Symbol is a gap with a lever (1).	1+1+1
	OR: Filament heated to glow (1); LED produces light directly (1); Less energy as heat (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Join + to - in series (1); 3 cells: 4.5 V (1); 5 cells: 7.5 V (1); Bulb brighter (1); Excess voltage can fuse a bulb (1).	1+1+1+1+1
	OR: Different appliances need different current (1); Rating suits the appliance (1); Protects from overload (1); Prevents false cut-offs (1); Safety (1).	1+1+1+1+1
20	Filament bulb: much heat (1); Filament bulb uses more energy (1); LED: least heat (1); LED lasts longer (1); Choose LED (1).	1+1+1+1+1
	OR: Cell to switch (1); Switch to fuse (1); Fuse to bulb (1); Bulb back to cell (1); If the fuse blows, the circuit breaks (1).	1+1+1+1+1

Mistakes to avoid:

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